

Phase 1: Project Structure and Configuration Files

Now that your environment is set up, let's lay the groundwork for your Autonomous Financial Research Agent by creating the necessary project structure and configuration files. This ensures your project is organized, maintainable, and ready for development.

Step 1: Create Your Project Directory

If you haven't already, create a dedicated folder for your project. Let's assume you named it `autonomous_financial_agent`.

Bash

```
mkdir autonomous_financial_agent
cd autonomous_financial_agent
```

Step 2: Initialize a Git Repository

Version control is essential. Initialize a Git repository in your project folder and make your first commit.

Bash

```
git init
echo "# Autonomous Financial Research Agent" > README.md
git add README.md
git commit -m "Initial commit: Add README.md"
```

Then, create a new private repository on GitHub (e.g., `autonomous-financial-agent`) and link your local repository to it. Remember to keep your project private if you're going to include API keys in your `.env` file (which you should!).

Bash

```
git remote add origin https://github.com/YOUR_GITHUB_USERNAME/autonomous-financial-agent
git branch -M main
git push -u origin main
```

Step 3: Set Up a Python Virtual Environment

A virtual environment isolates your project's dependencies from other Python projects on your system. This is a best practice.

Bash

```
python3 -m venv venv
source venv/bin/activate # On Windows, use `venv\Scripts\activate`
```

You should see `(venv)` prefixing your terminal prompt, indicating the virtual environment is active.

Step 4: Install Core Dependencies

Install the initial Python libraries required for managing environment variables and for LangChain.

Bash

```
pip install python-dotenv langchain-core langchain-community
```

Step 5: Create the `.env` File for API Keys

This file will securely store your API keys. **Crucially, you must add `.env` to your `.gitignore` file so it's never committed to GitHub.**

First, create the `.gitignore` file:

Bash

```
echo "venv/" > .gitignore
echo ".env" >> .gitignore
git add .gitignore
git commit -m "Add .gitignore to exclude venv and .env"
```

Now, create the `.env` file and add your API keys from Groq and Serper.dev. Replace `YOUR_GROQ_API_KEY` and `YOUR_SERPER_API_KEY` with the actual keys you obtained during the setup phase.

Python

```
# .env file
GROQ_API_KEY="YOUR_GROQ_API_KEY"
SERPER_API_KEY="YOUR_SERPER_API_KEY"
```

Step 6: Create `main.py` and `tools.py`

These will be your main Python files. `main.py` will orchestrate the agent, and `tools.py` will house the custom tools your agent uses.

Bash

```
touch main.py
touch tools.py
```

Step 7: Verify Environment Variables Loading

Add the following code to your `main.py` to ensure your environment variables are loading correctly. Run this file to confirm.

Python

```
# main.py

import os
from dotenv import load_dotenv

# Load environment variables from .env file
load_dotenv()

# Access your API keys
groq_api_key = os.getenv("GROQ_API_KEY")
serper_api_key = os.getenv("SERPER_API_KEY")

if groq_api_key and serper_api_key:
    print("API keys loaded successfully!")
else:
    print("Error: API keys not loaded. Check your .env file.")

# You can print them to verify, but remove this line before sharing code
# print(f"Groq Key: {groq_api_key[:5]}...")
# print(f"Serper Key: {serper_api_key[:5]}...")
```

Run `python main.py` in your terminal (with your virtual environment activated). You should see "API keys loaded successfully!" If not, double-check your `.env` file and ensure `load_dotenv()` is called.

With these steps completed, your project structure is solid, and your environment is configured. You are now ready to start developing the custom tools for your agent.